

Characterization of Various Impairments in Coherent Optical Transmitters Using Spectrum Analyser Only

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Abstract *Using probability-maintained multi-notch sequences and equally spaced multi-tone signals, various impairments in EDFA-embedded coherent transmitters including nonlinear distortion, additive electrical noise, and ASE noise are decoupled via spectrum-analyser-based measurements. The system's error vector magnitude is predicted with 0.4-dB accuracy based on the measured noise-to-signal ratio. ©2026 The Author(s)*

Introduction

Noise characterization and performance evaluation are important for designing and manufacturing transmitters [1,2]. Transmitter performance metrics obtained using golden receivers embedded with digital signal processing (DSP), such as transmitter dispersion eye closure quaternary [3] and extended transmitter constellation closure [4], are widely used as the overall performance margin. However, there are various impairment sources, such as in-phase/quadrature (IQ) imbalance, additive electrical noise, device nonlinear distortion, and amplified spontaneous emission (ASE) noise, in the coherent transmitter embedded with a booster Erbium-doped fibre amplifier (EDFA). Those metrics based on golden receivers cannot distinguish different impairment sources. Furthermore, their accuracy is affected by receiver noise [5] and DSP configuration [6,7]. Therefore, it is desirable to find a new approach to separate different impairment sources without impacts from receiver and DSP.

Spectrum analysis provides a receiver- and DSP-free solution for transmitter evaluation. By designing different test stimuli, various impairments could be characterized. For example, IQ imbalance could be characterized using double- and single-side spectrum notches [8]. A probability-maintained (PM) notch can accurately measure the nonlinear distortion when additive noise is ignorable [9,10]. In-band ASE noise can be estimated using out-of-band ASE noise with interpolation [11]. However, when various impairments coexist, it is challenging to separate them and to precisely evaluate transmitter performance [12].

In this paper, a spectrum-analyser-based method is proposed to characterize the impairments and estimate the performance of a coherent transmitter, in which the PM multi-notch sequence [13] and equally spaced multi-tone signal are specifically designed to separate the additive noise and nonlinear distortion. The noise-to-

signal ratios (NSRs) arising from transmitter nonlinear distortion, additive electrical noise, and ASE noise are decoupled. The total NSR, as a performance metric, achieves 0.4 dB accuracy compared to the error vector magnitude (EVM) obtained from the conventional receiver method.

Principle of Spectrum-Analyser-Based Impairment Characterization

Fig. 1(a) illustrates the model of an EDFA-embedded coherent transmitter, which consists of various impairments including transmitter additive electrical noise, nonlinear distortion, and ASE noise. Two stimuli are specifically designed for characterizing these impairments by spectrum-analyser-based measurement.

The first stimulus is the PM notch sequence, which is used to measure all the noises in the notch region. Its power spectral density (PSD) has a deep notch as shown in Fig. 1(b1) and its probability density function is the same as that of a standard quadrature amplitude modulation (QAM) sequence, guaranteeing the accurate nonlinear distortion measurement compared with the simple notch sequence [9,10]. The PM notch sequence is impaired by transmitter additive electrical noise and nonlinear distortion at observation point 2 as shown in Fig. 1(b2). As a booster EDFA is integrated with the coherent transmitter to amplify the output optical signal, the PM notch sequence is then impaired by ASE noise at observation point 3 as shown in Fig. 1(b3).

The second stimulus is the equally spaced multi-tone signal, whose tones are uniformly distributed across the signal bandwidth as shown in Fig. 1(c1). It is used to separate nonlinear distortion and additive noise. The principle behind this separation is that when the nonlinear system input has equally spaced frequency characteristics, the generated nonlinear distortion occurs at the same frequency grid as long as the device nonlinear behaviour is governed by Volterra series

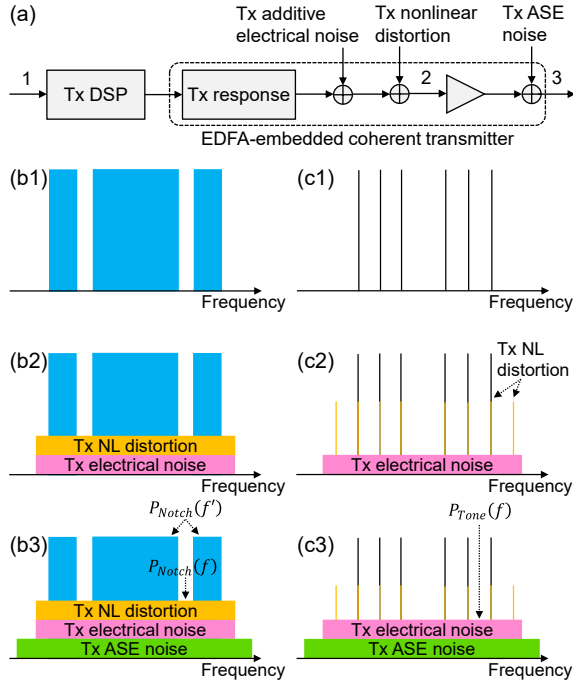


Fig. 1: (a) Model of EDFA-embedded coherent transmitter and schematic of noise spectrum measurement using (b1)/(b2)/(b3) probability-maintained notch sequence and (c1)/(c2)/(c3) equally spaced multi-tone signal at observation point 1/2/3.

[14,15], e.g., electrical driver in transmitter [16]. As a result, the nonlinear distortion has a discrete spectrum and does not appear between the tones as shown in Fig. 1(c2). It means the frequency components between two adjacent tones consist solely of the transmitter additive electrical noise. Thus, the transmitter nonlinear distortion and additive electrical noise are separated. After EDFA, the equally spaced multi-tone signal is further impaired by ASE noise as shown in Fig. 1(c3).

With the spectrum measured using PM notch sequence at observation point 3 described by P_{Notch} , the NSR including all transmitter noises can be obtained by

$$NSR_{Tx}(f) = \frac{P_{Notch}(f)}{P_{Notch}(f') - P_{Notch}(f)} \quad (1)$$

where $P_{Notch}(f)$ denotes the average power of all transmitter noises in the notch region and $P_{Notch}(f')$ denotes the average power of the signal and all transmitter noises in the regions adjacent to the notch. Due to limited depth of PM notch sequence, the calibration must be implemented to obtain the accurate NSR [9,10].

The contribution of each noise to system performance degradation could be characterized separately. The transmitter ASE noise level is

$$NSR_{Tx ASE}(f) = \frac{P_{ASE}(f)}{P_{Notch}(f') - P_{Notch}(f)} \quad (2)$$

where $P_{ASE}(f)$ denotes the average power of the in-band ASE noise in the notch region obtained by out-of-band ASE noise with interpolation [11]. Then, the performance degradation caused by

transmitter impairments excluding ASE noise is

$$NSR'_{Tx}(f) = NSR_{Tx}(f) - NSR_{Tx ASE}(f) \quad (3)$$

With the spectrum measured using equally spaced multi-tone signal at observation point 3 described by P_{Tone} , the level of transmitter additive electrical noise is obtained by

$$NSR_{Tx EleNoise}(f) = \frac{P_{Tone}(f) - P_{ASE}(f)}{P_{Notch}(f') - P_{Notch}(f)} \quad (4)$$

where $P_{Tone}(f)$ denotes the average power of the transmitter additive electrical noise and ASE noise in the corresponding notch region. Finally, the transmitter nonlinear distortion level is

$$NSR_{Tx NL}(f) = NSR'_{Tx}(f) - NSR_{Tx EleNoise}(f) \quad (5)$$

Note that these NSRs could be measured at different notch centre frequencies, and their averages are considered as indicators for transmitter performance estimation and noise analysis.

Compared to the conventional receiver method, the proposed method only requires a spectrum analyser, enabling transmitter performance evaluation without the impacts of receiver noise [5] and DSP configuration [6,7]. The sole requirement for the proposed method is a sufficient resolution bandwidth (RBW) of the spectrum analyser.

Experimental Setup and Results

Fig. 2 shows the experimental setup and DSP flow. Three dual-polarization signals are used in experiment. The standard 64 Gbaud 64QAM is used for performance evaluation via conventional receiver method as a reference. All parameters of receiver DSP are optimized to obtain reliable EVMs. The 64QAM PM multi-notch sequence was generated for fast and accurate noise characterization and performance evaluation [13], whose PSD has double-side 6 notches at $\pm 5.33/16/26.67$ GHz with 27 dB notch depth and 1.6 GHz notch width for each notch. The equally spaced multi-tone signal has double-side 18 tones with 3.56 GHz frequency interval. Note that these signals have the same root-mean-square amplitude in transmitter DSP. Nonlinear distortion level was changed by adjusting the gain of electrical driver in high bandwidth coherent driver modulator (HB-CDM). ASE noise level was varied by adjusting the EDFA input power using a variable optical attenuator (VOA) 1. The receiver input power was fixed by adjusting VOA2 to keep the impact of receiver noise on system performance unchanged significantly. These powers were monitored by an optical power meter. A high-resolution optical spectrum analyser (HR-OSA) with a 150 MHz RBW measured the HB-CDM and EDFA output spectra and thus NSRs.

Fig. 3(a) depicts the normalized spectra of equally spaced multi-tone signal measured at observation point 2 and different nonlinear conditions. The generated nonlinear distortion has the same frequency characteristics with the equally

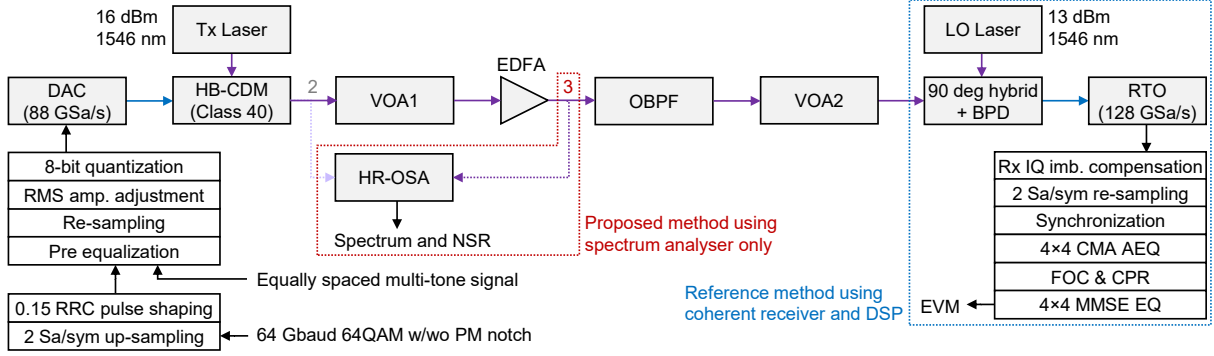


Fig. 2: Experimental setup and DSP flow.

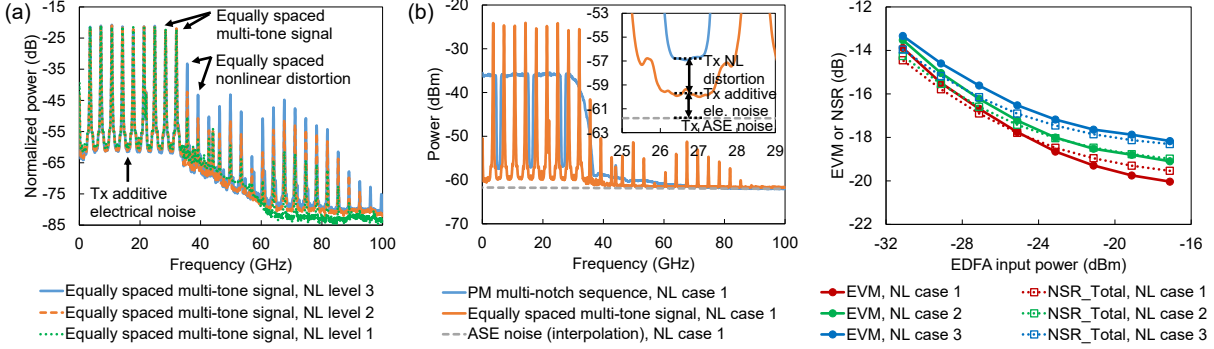


Fig. 3: Spectra measured at (a) observation point 2 and (b) observation point 3.

spaced multi-tone signal at different nonlinear distortion levels, which verifies the separation of the transmitter additive electrical noise and nonlinear distortion. Fig. 3(b) depicts the spectra of PM multi-notch sequence and equally spaced multi-tone signal measured at observation point 3. The inset shows the details of the measured spectra around the 3rd notch region, demonstrating the necessity and feasibility of separating the transmitter nonlinear distortion, additive electrical noise, and ASE noise with the proposed method.

To validate the proposed transmitter performance evaluation method, $NSR_{Total} = NSR_{Tx} + NSR_{Rx}$ was compared with the EVM obtained using conventional receiver method, where NSR_{Rx} was -21.9 dB determined by offline calibration on the output of the real-time oscilloscope. As shown in Fig. 4, NSR_{Total} agrees with EVM at different transmitter nonlinear distortion levels and different EDFA input power levels (and thus different optical signal-to-noise ratios), achieving 0.4 dB root mean square error (RMSE). Fig. 5 shows the separation of NSR_{Tx} into three components, $NSR_{Tx ASE}$ as the ASE noise level, $NSR_{Tx NL}$ as the transmitter nonlinear distortion level, and $NSR_{Tx EleNoise}$ as the additive electrical noise level, using the proposed method. Each component exhibits a distinct behaviour. Specifically, $NSR_{Tx ASE}$ depends solely on the EDFA input power, irrespective of the transmitter nonlinear distortion level (three dashed curves are overlapped). In contrast, $NSR_{Tx NL}$ is governed by the transmitter nonlinear distortion level and is independent of the EDFA input power. $NSR_{Tx EleNoise}$

Fig. 4: EVM and NSR_{Total} .

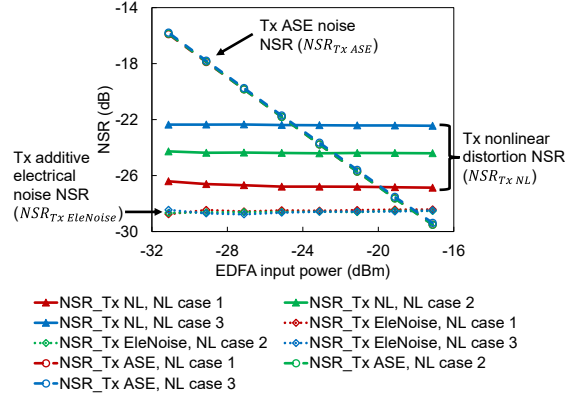


Fig. 5: Results of NSR separation.

remains invariant to changes in both transmitter nonlinear distortion and EDFA input power (three dotted curves are overlapped). The observed behaviours of the NSR for each impairment under different system setups are highly consistent with theoretical expectations, validating the effectiveness of the proposed method.

Conclusions

Using the specifically designed PM multi-notch sequences and equally spaced multi-tone signals, the performance of a coherent transmitter with an EDFA was evaluated by a HR-OSA, achieving a 0.4 dB RMSE compared to the conventional receiver method. The performance degradations caused by transmitter nonlinear distortion, additive electrical noise, and ASE noise were estimated separately, providing valuable insights for impairment source analysis and system design.

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